DNS e HTTP con router

Setup

```
cd /tmp
git clone https://github.com/Marrocco-Simone/reti-lab-kathara
cd reti-lab-kathara/lab5-dns-http-routed
./build-images.sh
```

Attenzione: `/tmp` viene cancellata allo spegnimento del PC. Copiate altrove i file che modificate!

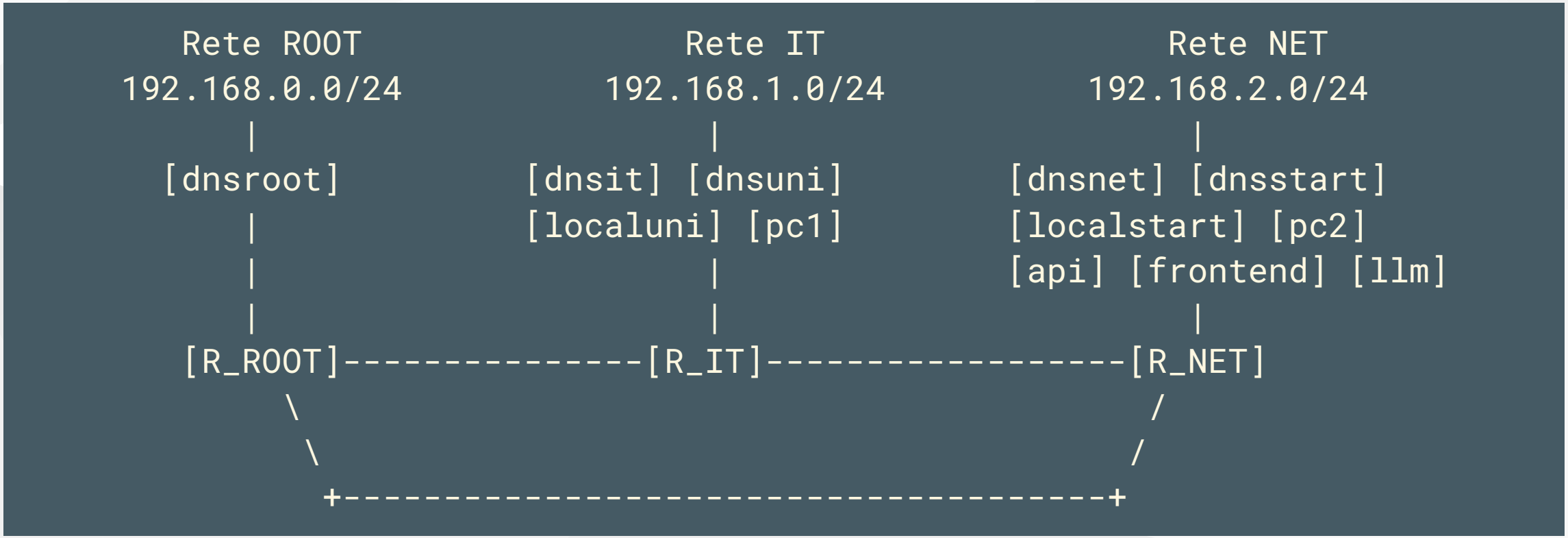
Esercizio: Separare i Domini DNS

Attualmente tutti i nodi sono nella stessa rete "A".

Vogliamo separare i tre domini principali:

- **Dominio ROOT** (.): dnsroot
- **Dominio .it**: dnsit, dnsuni, localuni, pc1
- **Dominio .net**: dnsnet, dnsstart, localstart, pc2, api, frontend, llm

Topologia Target



I tre router sono tutti connessi tra loro (topologia a triangolo).

I Tre Router

Router	eth0	eth1	eth2
R_ROOT	ROOT (192.168.0.0/24)	Link a R_IT	Link a R_NET
R_IT	IT (192.168.1.0/24)	Link da R_ROOT	Link a R_NET
R_NET	NET (192.168.2.0/24)	Link da R_IT	Link da R_ROOT

I Tre Router

Reti di collegamento tra i router:

- **LINK_RI** (R_ROOT ↔ R_IT): 192.168.10.0/24
- **LINK_IN** (R_IT ↔ R_NET): 192.168.20.0/24
- **LINK_RN** (R_ROOT ↔ R_NET): 192.168.30.0/24

Modifiche a lab.conf (1/2)

```
# Rete ROOT - dominio radice
dnsroot[0]="ROOT"

# Rete IT - dominio .it
dnsit[0]="IT"
dnsuni[0]="IT"
localuni[0]="IT"
pc1[0]="IT"

# Rete NET - dominio .net
dnsnet[0]="NET"
dnsstart[0]="NET"
localstart[0]="NET"
pc2[0]="NET"
api[0]="NET"
frontend[0]="NET"
llm[0]="NET"
```

Modifiche a lab.conf (2/2)

```
# Router ROOT: connette ROOT, R_IT e R_NET
r_root[0]="ROOT"        # eth0 -> rete ROOT
r_root[1]="LINK_RI"     # eth1 -> link verso R_IT
r_root[2]="LINK_RN"     # eth2 -> link verso R_NET

# Router IT: connette IT, R_ROOT e R_NET
r_it[0]="IT"            # eth0 -> rete IT
r_it[1]="LINK_RI"       # eth1 -> link da R_ROOT
r_it[2]="LINK_IN"       # eth2 -> link verso R_NET

# Router NET: connette NET, R_IT e R_ROOT
r_net[0]="NET"          # eth0 -> rete NET
r_net[1]="LINK_IN"      # eth1 -> link da R_IT
r_net[2]="LINK_RN"      # eth2 -> link da R_ROOT
```

Schema delle Reti

Nome Rete	CIDR	Contenuto
ROOT	192.168.0.0/24	dnsroot, R_ROOT
IT	192.168.1.0/24	dnsit, dnsuni, localuni, pc1, R_IT
NET	192.168.2.0/24	dnsnet, dnsstart, ..., R_NET
LINK_RI	192.168.10.0/24	R_ROOT ↔ R_IT
LINK_IN	192.168.20.0/24	R_IT ↔ R_NET
LINK_RN	192.168.30.0/24	R_ROOT ↔ R_NET

Indirizzi dei Router

```
R_ROOT:
  eth0: 192.168.0.100/24   (rete ROOT)
  eth1: 192.168.10.1/24    (LINK_RI verso R_IT)
  eth2: 192.168.30.1/24    (LINK_RN verso R_NET)

R_IT:
  eth0: 192.168.1.100/24   (rete IT)
  eth1: 192.168.10.2/24    (LINK_RI da R_ROOT)
  eth2: 192.168.20.1/24    (LINK_IN verso R_NET)

R_NET:
  eth0: 192.168.2.100/24   (rete NET)
  eth1: 192.168.20.2/24    (LINK_IN da R_IT)
  eth2: 192.168.30.2/24    (LINK_RN da R_ROOT)
```

Nuovi Indirizzi dei Server DNS

Server	Vecchio IP	Nuovo IP
dnsroot	192.168.0.5	192.168.0.5
dnsit	192.168.0.1	192.168.1.1
dnsuni	192.168.0.11	192.168.1.11
dnsnet	192.168.0.2	192.168.2.2
dnsstart	192.168.0.22	192.168.2.22

Attenzione: dovrete aggiornare anche i file di zona DNS!

File .startup dei Router

```
# r_root.startup
ip a add 192.168.0.100/24 dev eth0
ip a add 192.168.10.1/24 dev eth1
ip a add 192.168.30.1/24 dev eth2
ip link set eth0 up
ip link set eth1 up
ip link set eth2 up

# Route verso IT (via LINK_RI) e NET (via LINK_RN)
ip route add 192.168.1.0/24 via 192.168.10.2   # rete IT via R_IT
ip route add 192.168.2.0/24 via 192.168.30.2   # rete NET via R_NET
ip route add 192.168.20.0/24 via 192.168.10.2  # LINK_IN via R_IT
```

File .startup dei Router

```
# r_it.startup
ip a add 192.168.1.100/24 dev eth0
ip a add 192.168.10.2/24 dev eth1
ip a add 192.168.20.1/24 dev eth2
ip link set eth0 up
ip link set eth1 up
ip link set eth2 up

# Route verso ROOT (via LINK_RI) e NET (via LINK_IN)
ip route add 192.168.0.0/24 via 192.168.10.1   # rete ROOT via R_ROOT
ip route add 192.168.2.0/24 via 192.168.20.2   # rete NET via R_NET
ip route add 192.168.30.0/24 via 192.168.10.1  # LINK_RN via R_ROOT
```

File .startup dei Router

```
# r_net.startup
ip a add 192.168.2.100/24 dev eth0
ip a add 192.168.20.2/24 dev eth1
ip a add 192.168.30.2/24 dev eth2
ip link set eth0 up
ip link set eth1 up
ip link set eth2 up

# Route verso ROOT (via LINK_RN) e IT (via LINK_IN)
ip route add 192.168.0.0/24 via 192.168.30.1   # rete ROOT via R_ROOT
ip route add 192.168.1.0/24 via 192.168.20.1   # rete IT via R_IT
ip route add 192.168.10.0/24 via 192.168.20.1  # LINK_RI via R_IT
```

Modifiche ai .startup dei Server

Ogni server deve:

1. Cambiare il proprio IP
2. Aggiungere il default gateway

```
# Esempio: dnsroot.startup
ip address add 192.168.0.5/24 dev eth0
ip link set eth0 up
route add default gw 192.168.0.100 dev eth0
systemctl start named
```

Modifiche ai .startup dei Client

```
# pc1.startup (rete IT)
ip address add 192.168.1.111/24 dev eth0
ip link set eth0 up
route add default gw 192.168.1.100 dev eth0

# pc2.startup (rete NET)
ip address add 192.168.2.222/24 dev eth0
ip link set eth0 up
route add default gw 192.168.2.100 dev eth0
```

Aggiornare i File di Zona DNS

I file in `dnsroot/etc/bind/db.root` devono riflettere i nuovi IP:

```
# Prima
dnsit.it.       IN    A    192.168.0.1
dnsnet.net.     IN    A    192.168.0.2

# Dopo
dnsit.it.       IN    A    192.168.1.1
dnsnet.net.     IN    A    192.168.2.2
```

Lo stesso per tutti gli altri file di zona!

Far partire Kathara

```
# nella cartella /tmp/reti-lab-kathara/lab5-dns-http-routed
./start-lab.sh
kathara connect pc1
```

```
traceroute 192.168.2.112
dig A api.startup.net
curl http://api.startup.net/visitors
```

Da quali router passano i pacchetti? Quale rete attraversa la query DNS?

Wireshark con più reti

Con i router, Wireshark viene connesso a **tutte le reti**:

Interfaccia	Rete	Cosa vedi
eth1	ROOT	Traffico verso dnsroot
eth2	IT	Traffico di pc1, dnsit, dnsuni
eth3	NET	Traffico di pc2, api, frontend
eth4	LINK_RI	Pacchetti tra R_ROOT e R_IT
eth5	LINK_IN	Pacchetti tra R_IT e R_NET
eth6	LINK_RN	Pacchetti tra R_ROOT e R_NET